

The rank-corner formula at $r = 2$:
 $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2$ for $\lambda = (3, 2, 1^{n-5})$

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Abstract

We prove that $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$ for every $n \geq 8$ at $\lambda = (3, 2, 1^{n-5})$. This closes the dimension gap left open in the May-20 paper `2026-05-20-3-2-1n-family.tex`, which proved E_1^- containment (hence rank-zero) but only the upper bound $\dim \leq 2$. The lower bound is the **first proven instance of the rank-corner formula at $r = 2$ length-preserving corners**.

The argument is a *position-of- n invariant*: the operators T_j for $j \leq n - 2$ preserve the position of the letter n in every SYT. So if $\pi^{(c)} : V_\lambda \rightarrow V_\lambda$ is the seminormal-basis projection onto SYTs with n at the cell c , then $\pi^{(c)}$ commutes with each (T_j+1) for $j \leq n - 2$. Under the May-20 three-branch reduction $B_{n-2}^{(\lambda)} V_\lambda = \langle F_A \rangle + \langle F_B \rangle$, the A -piece F_A has support only on SYTs with $n \in \{(1, 3), (n - 3, 1)\}$, and the B -piece F_B has support on $\{(2, 2), (n - 3, 1)\}$. The cells $(1, 3)$ and $(2, 2)$ are exclusive, so it suffices to show $\pi^{(1,3)} F_A \neq 0$ and $\pi^{(2,2)} F_B \neq 0$. A four-line Hoefsmit case analysis on the μ_X -supports does each.

Verification. $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$ verified at $q = 7$ for $n \in \{8, 9, 10\}$; the exclusive $\pi^{(1,3)}$ -witnesses (4 SYTs) and $\pi^{(2,2)}$ -witnesses (5 SYTs) match the case-analysis prediction exactly.

1 Setup and statement

Notation as in `2026-05-20-3-2-1n-family.tex` (henceforth **May-20**). Throughout, $n \geq 8$ and

$$\lambda := (3, 2, 1^{n-5}) \vdash n, \quad \ell := \ell(\lambda) = n - 3, \quad j_0 := \ell + 1 = n - 2.$$

The three corners of λ are $c_1 = (1, 3)$, $c_2 = (2, 2)$ (both length-preserving), and $c_3 = (n - 3, 1)$ (column-1 bottom). For each pair $\{c_i, c_j\}$, the doubly-stripped partition is

$$\mu_A := (2, 2, 1^{n-6}) \vdash n - 2, \quad \mu_B := (3, 1^{n-5}) \vdash n - 2, \quad \mu_C := (2, 1^{n-4}) \vdash n - 2,$$

indexed so μ_A omits $\{c_1, c_3\}$, μ_B omits $\{c_2, c_3\}$, and μ_C omits $\{c_1, c_2\}$.

We rely on three external results:

Theorem 1 (May-20 three-branch reduction, Theorem 3.5).

$$B_{n-2}^{(\lambda)} V_\lambda = (T_{n-3}+1)(T_{n-2}+1) (\Phi_A(u^{(\mu_A)}) + \Phi_B(u^{(\mu_B)})),$$

where $u^{(\mu_X)}$ spans $B_{n-3}^{(\mu_X)} V_{\mu_X}$ and the μ_C -piece vanishes via May-20 Lemma 4.1.

Theorem 2 (Support of $u^{(\mu_A)}$; May-13 Theorem 2.5 + May-18). $u^{(\mu_A)}$ has seminormal support exactly

$$S_A := \{(n-5, n-4), (n-5, n-3), (n-4, n-2), (n-3, n-2)\}$$

in the (α, γ) -indexing of May-13 (column-2 entries of μ_A), with all four coefficients in $\mathbb{Q}(q)^\times$.

Theorem 3 (Support of $u^{(\mu_B)}$; May-12 Proposition 6.5). $u^{(\mu_B)}$ has seminormal support exactly

$$S_B := \{(n-5, n-4), (n-5, n-3), (n-4, n-3), (n-4, n-2), (n-3, n-2)\}$$

in the (a, b) -indexing of May-12 (row-1 entries at $(1, 2), (1, 3)$ of μ_B), with all five coefficients in $\mathbb{Q}(q)^\times$.

The main result of this paper:

Theorem 4 (Rank-corner formula at $r = 2$). For every $n \geq 8$, $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$. Equivalently, the rank of $B_{\ell+1}^{(\lambda)}$ on V_λ equals the number of length-preserving corners of λ .

May-20 Theorem 6.4 gives $\dim \leq 2$; this paper proves $\dim \geq 2$.

2 The position-of- n invariant

Lemma 5 (Position of n is invariant under T_j for $j \leq n-2$). For every SYT T of λ and every $1 \leq j \leq n-2$, the position $\text{pos}_T(n)$ of the letter n is unchanged under the action of T_j on v_T . More precisely, $T_j v_T$ is a $\mathbb{Q}(q)$ -linear combination of seminormal basis vectors $v_{T'}$ with $\text{pos}_{T'}(n) = \text{pos}_T(n)$.

Proof. T_j acts in the seminormal basis by the standard formula: same row $\Rightarrow q$, same column $\Rightarrow -1$, otherwise a Hoefsmit 2-block on $\{v_T, v_{s_j T}\}$ where $s_j T$ swaps the letters $j, j+1$. In all three cases, $T_j v_T$ is supported on SYTs related to T by either the identity or the swap of j and $j+1$. Since $j+1 \leq n-1$, neither swap touches the letter n , so $\text{pos}(n)$ is preserved. $\square$

Corollary 6 (Position- n projections commute with (T_j+1)). For each cell c of λ , define the seminormal-basis projection

$$\pi^{(c)} : V_\lambda \rightarrow V_\lambda, \quad \pi^{(c)} v_T = \begin{cases} v_T & \text{pos}_T(n) = c \\ 0 & \text{otherwise.} \end{cases}$$

Then $\pi^{(c)}$ commutes with (T_j+1) for every $1 \leq j \leq n-2$.

Proof. Immediate from Lemma 5, since T_j preserves each summand $V_\lambda = \bigoplus_c \text{im}(\pi^{(c)})$, and hence so does $T_j + 1$. $\square$

3 Support of F_A and F_B under $\pi^{(c)}$

Define

$$F_A := (T_{n-3}+1)(T_{n-2}+1) \Phi_A(u^{(\mu_A)}), \quad F_B := (T_{n-3}+1)(T_{n-2}+1) \Phi_B(u^{(\mu_B)}).$$

By Theorem 1, $B_{n-2}^{(\lambda)} V_\lambda = \mathbb{Q}(q) F_A + \mathbb{Q}(q) F_B$.

Lemma 7 (n -position support of F_A and F_B). Every SYT $T \in \text{supp}(F_A)$ has $\text{pos}_T(n) \in \{c_1, c_3\} = \{(1, 3), (n-3, 1)\}$. Every SYT $T \in \text{supp}(F_B)$ has $\text{pos}_T(n) \in \{c_2, c_3\} = \{(2, 2), (n-3, 1)\}$.

Proof. By construction of Φ_A (May-20 Definition 2.2), every SYT in the seminormal support of $\Phi_A(V_{\mu_A})$ has $\{n-1, n\}$ occupying the cell pair $\{c_1, c_3\}$ in one of the two orderings. In particular, $\text{pos}(n) \in \{c_1, c_3\}$ for every such SYT. By Corollary 6, this is preserved under $(T_{n-3}+1)(T_{n-2}+1)$. The same argument applies to Φ_B with $\{c_2, c_3\}$. $\square$

Corollary 8 (Cross-projection vanishings). $\pi^{(c_1)}F_B = 0$ and $\pi^{(c_2)}F_A = 0$.

Proof. F_B 's support has $\text{pos}(n) \in \{c_2, c_3\}$, never c_1 . Similarly F_A 's support never has $\text{pos}(n) = c_2$. $\square$

4 $\pi^{(c_1)}F_A \neq 0$ (Lemma A)

In this section we identify $V_\lambda|_{\text{pos}(n)=c_1}$ with the seminormal module V_{λ^-} for $\lambda^- := \lambda \setminus c_1 = (2, 2, 1^{n-5}) \vdash n-1$ as $H_q(S_{n-1})$ -modules: each SYT T of λ with n at $(1, 3)$ corresponds to the SYT T' of λ^- on letters $1, \dots, n-1$ obtained by deleting the cell $(1, 3)$.

For each $(\alpha, \gamma) \in S_A$, write $T_{(\alpha, \gamma)}^{A, B}$ for the SYT of λ obtained from the μ_A -SYT with column-2 entries (α, γ) by Φ_A in the “ n at c_1 ” direction — i.e., n at $(1, 3)$, $n-1$ at $(n-3, 1)$.

We use the Hoefsmit 2-block in the form: for T_i acting on $\{v_T, v_{s_i T}\}$ at axial distance $d = c(i+1) - c(i)$ with $|d| \geq 2$, write

$$T_i v_T = a_d v_T + b_d v_{s_i T}, \quad (T_i + 1) v_T = (a_d + 1) v_T + b_d v_{s_i T}. \quad (1)$$

The eigenvalues of T_i on this 2-block are $\{q, -1\}$, so the trace gives $a_d + a'_d = q - 1$ (where a'_d is the diagonal entry on $v_{s_i T}$) and the determinant gives $a_d a'_d - b_d b'_d = -q$. *Non-degeneracy* of the 2-block (i.e., $b_d \neq 0$ and $b'_d \neq 0$) holds iff $|d| \neq 1$ (see May-13 §2.2 / Murphy's seminormal basis); under non-degeneracy, $a_d \neq -1$ as well, since $a_d = -1$ would force $a'_d = q$ and then $a_d a'_d = -q$, requiring $b_d b'_d = 0$, a contradiction. Hence for $|d| \geq 2$ both $a_d + 1 \in \mathbb{Q}(q)^\times$ and $b_d \in \mathbb{Q}(q)^\times$.

4.1 Case analysis on S_A

We decompose $\pi^{(c_1)}F_A$ as a sum over the four supports of $u^{(\mu_A)}$. Since $\Phi_A(v_{T'}^{(\mu_A)}) = v_{T_{(\alpha, \gamma)}^{A, A}} + \tau_A v_{T_{(\alpha, \gamma)}^{A, B}}$ with $\tau_A \in \mathbb{Q}(q)^\times$ (May-20 Definition 2.2, axial distance $|d_A| = n-2 \geq 6$ at $n \geq 8$), the c_1 -projection selects the τ_A piece:

$$\pi^{(c_1)}\Phi_A(u^{(\mu_A)}) = \tau_A \sum_{(\alpha, \gamma) \in S_A} A_{(\alpha, \gamma)} v_{T_{(\alpha, \gamma)}^{A, B}},$$

where $A_{(\alpha, \gamma)} \in \mathbb{Q}(q)^\times$ are the coefficients of $u^{(\mu_A)}$ (Theorem 2).

For each $(\alpha, \gamma) \in S_A$ we compute $(T_{n-3}+1)(T_{n-2}+1)v_{T_{(\alpha, \gamma)}^{A, B}}$. The positions of letters $n-3, n-2$ in $T_{(\alpha, \gamma)}^{A, B}$ depend on the μ_A -data:

Positions in $T_{(\alpha, \gamma)}^{A, B}$. In any μ_A -SYT with column-2 entries (α, γ) on $n-2$ letters, the column-1 entries fill cells $(1, 1), (2, 1), \dots, (n-4, 1)$ with the elements of $\{1, \dots, n-2\} \setminus \{\alpha, \gamma\}$ in increasing order. Consequently:

- If $\gamma = n-2$: the cell $(2, 2) = n-2$ in T' . The maximum column-1 entry is $\alpha-1$ if $\alpha = n-3$, else $n-3$.
- If $\gamma < n-2$: the cell $(2, 2) = \gamma$. The maximum column-1 entry is $n-2$ (at $(n-4, 1)$).

For the four $(\alpha, \gamma) \in S_A$:

(α, γ)	$\text{pos}_{T'}(n-2)$	$\text{pos}_{T'}(n-3)$	$\text{pos}(n-1)$ in $T^{A, B}$	$\text{pos}(n)$ in $T^{A, B}$
$(n-5, n-4)$	$(n-4, 1)$	$(n-5, 1)$	$(n-3, 1)$	$(1, 3)$
$(n-5, n-3)$	$(n-4, 1)$	$(2, 2) (= \gamma)$	$(n-3, 1)$	$(1, 3)$
$(n-4, n-2)$	$(2, 2)$	$(n-4, 1)$	$(n-3, 1)$	$(1, 3)$
$(n-3, n-2)$	$(2, 2)$	$(1, 2) (= \alpha)$	$(n-3, 1)$	$(1, 3)$

We compute the action of $(T_{n-3}+1)(T_{n-2}+1)$ on each $v_{T_{(\alpha,\gamma)}^{A,B}}$:

(A1) $(\alpha, \gamma) = (n-5, n-4)$. $\text{pos}(n-2) = (n-4, 1)$, $\text{pos}(n-1) = (n-3, 1)$ are in the same column, adjacent. Thus $T_{n-2}v = -v$ and

$$(T_{n-2}+1)v_{T_{(n-5,n-4)}^{A,B}} = 0.$$

Contribution to $\pi^{(c_1)}F_A$: **zero**.

(A2) $(\alpha, \gamma) = (n-5, n-3)$. Same analysis as (A1): $\text{pos}(n-2) = (n-4, 1)$, $\text{pos}(n-1) = (n-3, 1)$, same column adjacent.

$$(T_{n-2}+1)v_{T_{(n-5,n-3)}^{A,B}} = 0.$$

Contribution: **zero**.

(A3) $(\alpha, \gamma) = (n-4, n-2)$. $\text{pos}(n-2) = (2, 2)$, $\text{pos}(n-1) = (n-3, 1)$: different row and column. T_{n-2} acts as a 2-block at axial distance $d_1 = c(n-3, 1) - c(2, 2) = (4-n) - 0 = 4-n$. For $n \geq 8$, $|d_1| \geq 4$.

By (1), $(T_{n-2}+1)v_{T_{(n-4,n-2)}^{A,B}} = (a_{d_1}+1)v_{T_{(n-4,n-2)}^{A,B}} + b_{d_1}v_{\sigma_2}$, where $\sigma_2 := s_{n-2}T_{(n-4,n-2)}^{A,B}$, obtained by swapping the letters $n-2$ and $n-1$. Explicitly, σ_2 has $\text{pos}(n-1) = (2, 2)$, $\text{pos}(n-2) = (n-3, 1)$. (Valid SYT: row 1 reads $(1, 1) < n-4 < n$; column 1 reads $1 < 2 < \dots < n-3 < n-2$ where the swap moved $n-2$ from $(2, 2)$ to $(n-3, 1)$ and shifted $n-3$ accordingly; column 2 reads $n-4 < n-1$.)

Now apply T_{n-3} :

- On $v_{T_{(n-4,n-2)}^{A,B}}$: $\text{pos}(n-3) = (n-4, 1)$, $\text{pos}(n-2) = (2, 2)$. 2-block at axial distance $d_2 = c(2, 2) - c(n-4, 1) = 0 - (5-n) = n-5$. For $n \geq 8$, $|d_2| \geq 3$. Hence

$$(T_{n-3}+1)v_{T_{(n-4,n-2)}^{A,B}} = (a_{d_2}+1)v_{T_{(n-4,n-2)}^{A,B}} + b_{d_2}v_{\sigma_3},$$

where $\sigma_3 := s_{n-3}T_{(n-4,n-2)}^{A,B}$: $\text{pos}(n-3) = (2, 2)$, $\text{pos}(n-2) = (n-4, 1)$.

- On v_{σ_2} : $\text{pos}(n-3) = (n-4, 1)$, $\text{pos}(n-2) = (n-3, 1)$. Same column, adjacent. $T_{n-3}v_{\sigma_2} = -v_{\sigma_2}$, so $(T_{n-3}+1)v_{\sigma_2} = 0$.

Combining,

$$(T_{n-3}+1)(T_{n-2}+1)v_{T_{(n-4,n-2)}^{A,B}} = (a_{d_1}+1)(a_{d_2}+1)v_{T_{(n-4,n-2)}^{A,B}} + (a_{d_1}+1)b_{d_2}v_{\sigma_3} + 0.$$

Both coefficients are in $\mathbb{Q}(q)^\times$.

(A4) $(\alpha, \gamma) = (n-3, n-2)$. $\text{pos}(n-3) = (1, 2) = \alpha$, $\text{pos}(n-2) = (2, 2) = \gamma$: same column, adjacent. Thus $T_{n-3}v_{T_{(n-3,n-2)}^{A,B}} = -v$, so applying $(T_{n-3}+1)$ to the “identity-piece” of the T_{n-2} 2-block yields zero.

$\text{pos}(n-2) = (2, 2)$ and $\text{pos}(n-1) = (n-3, 1)$: 2-block at axial distance $d_1 = 4-n$, same as in (A3). $(T_{n-2}+1)v_{T_{(n-3,n-2)}^{A,B}} = (a_{d_1}+1)v_{T_{(n-3,n-2)}^{A,B}} + b_{d_1}v_{\sigma'_2}$, where $\sigma'_2 := s_{n-2}T_{(n-3,n-2)}^{A,B}$: $\text{pos}(n-1) = (2, 2)$, $\text{pos}(n-2) = (n-3, 1)$.

Apply T_{n-3} :

- On $v_{T_{(n-3,n-2)}^{A,B}}$: same-column-2 pair $(1, 2), (2, 2)$, so $(T_{n-3}+1)v = 0$.

- On $v_{\sigma'_2}$: $\text{pos}(n-3) = (1, 2)$, $\text{pos}(n-2) = (n-3, 1)$: different row/column. 2-block at axial distance $d_3 = c(n-3, 1) - c(1, 2) = (4-n) - 1 = 3-n$, with $|d_3| \geq 5$ at $n \geq 8$. Thus $(T_{n-3}+1)v_{\sigma'_2} = (a_{d_3}+1)v_{\sigma'_2} + b_{d_3}v_{\sigma'_3}$, where $\sigma'_3 := s_{n-3}\sigma'_2$: $\text{pos}(n-3) = (n-3, 1)$, $\text{pos}(n-2) = (1, 2)$, $\text{pos}(n-1) = (2, 2)$.

Combining,

$$(T_{n-3}+1)(T_{n-2}+1)v_{T_{(n-3,n-2)}^{A,B}} = (a_{d_1}+1) \cdot 0 + b_{d_1} [(a_{d_3}+1)v_{\sigma'_2} + b_{d_3}v_{\sigma'_3}].$$

4.2 Disjointness of the four post-image supports

The four contributing supports of $\pi^{(c_1)}F_A$ are:

- From (A3): $\{T_{(n-4,n-2)}^{A,B}, \sigma_3\}$.
- From (A4): $\{\sigma'_2, \sigma'_3\}$.

We tabulate their column-2 entries $((1, 2), (2, 2))$:

SYT	(1, 2)	(2, 2)
$T_{(n-4,n-2)}^{A,B}$	$n-4$	$n-2$
$\sigma_3 = s_{n-3}T_{(n-4,n-2)}^{A,B}$	$n-4$	$n-3$
$\sigma'_2 = s_{n-2}T_{(n-3,n-2)}^{A,B}$	$n-3$	$n-1$
$\sigma'_3 = s_{n-3}\sigma'_2$	$n-2$	$n-1$

All four $((1, 2), (2, 2))$ values are distinct, so the four SYTs are distinct.

Proposition 9 (Lemma A). $\pi^{(c_1)}F_A \neq 0$. *Explicitly,*

$$\begin{aligned} \pi^{(c_1)}F_A = \tau_A [& A_{(n-4,n-2)}(a_{d_1}+1)(a_{d_2}+1)v_{T_{(n-4,n-2)}^{A,B}} + A_{(n-4,n-2)}(a_{d_1}+1)b_{d_2}v_{\sigma_3} \\ & + A_{(n-3,n-2)}b_{d_1}(a_{d_3}+1)v_{\sigma'_2} + A_{(n-3,n-2)}b_{d_1}b_{d_3}v_{\sigma'_3}], \end{aligned}$$

with all four coefficients in $\mathbb{Q}(q)^\times$.

Proof. The four contributing SYTs are distinct (table above) and each appears in $\pi^{(c_1)}F_A$ with coefficient the product of nonzero factors τ_A , the relevant $A_{(\alpha,\gamma)}$ (nonzero by Theorem 2), and Hoefsmit coefficients $a_{d_i}+1$ or b_{d_i} (nonzero by (1) since $|d_i| \geq 3$ in all cases at $n \geq 8$). Hence $\pi^{(c_1)}F_A$ has four nonzero seminormal coordinates. $\square$

5 $\pi^{(c_2)}F_B \neq 0$ (Lemma B)

The analysis for the B -piece is parallel, with the key difference that the μ_B -SYT in question is a hook $(3, 1^{n-5})$, indexed by the pair (a, b) at cells $(1, 2), (1, 3)$. We write $T_{(a,b)}^{B,B}$ for the SYT of λ obtained from the μ_B -SYT $T'_{(a,b)}$ by Φ_B in the “ n at c_2 ” direction: n at $(2, 2)$, $n-1$ at $(n-3, 1)$.

Positions in $T_{(a,b)}^{B,B}$. In $\mu_B = (3, 1^{n-5})$, the SYT $T'_{(a,b)}$ has 1 at $(1, 1)$, a at $(1, 2)$, b at $(1, 3)$, and column 1 entries $\{1, \dots, n-2\} \setminus \{a, b\}$ in increasing order at $(1, 1), (2, 1), \dots, (n-4, 1)$. Hence:

- If $b = n-2$: cell $(1, 3) = n-2$, and the maximum column-1 entry is $n-3$ if $a < n-3$, else $n-4$.

- If $b < n - 2$: cell $(1, 3) = b$, and the maximum column-1 entry is $n - 2$ at $(n - 4, 1)$.

For the five $(a, b) \in S_B$:

(a, b)	$\text{pos}(n - 2)$	$\text{pos}(n - 3)$	$\text{pos}(n - 1)$	$\text{pos}(n)$
$(n - 5, n - 4)$	$(n - 4, 1)$	$(n - 5, 1)$	$(n - 3, 1)$	$(2, 2)$
$(n - 5, n - 3)$	$(n - 4, 1)$	$(1, 3) (= b)$	$(n - 3, 1)$	$(2, 2)$
$(n - 4, n - 3)$	$(n - 4, 1)$	$(1, 3) (= b)$	$(n - 3, 1)$	$(2, 2)$
$(n - 4, n - 2)$	$(1, 3) (= b)$	$(n - 4, 1)$	$(n - 3, 1)$	$(2, 2)$
$(n - 3, n - 2)$	$(1, 3) (= b)$	$(1, 2) (= a)$	$(n - 3, 1)$	$(2, 2)$

(B1, B2, B3): $b \leq n - 3$. For $(a, b) \in \{(n - 5, n - 4), (n - 5, n - 3), (n - 4, n - 3)\}$, the position of $n - 2$ is $(n - 4, 1)$, same column as $\text{pos}(n - 1) = (n - 3, 1)$, adjacent. $(T_{n-2}+1)v_{T_{(a,b)}^{B,B}} = 0$. Contribution: **zero**.

(B4) $(a, b) = (n - 4, n - 2)$. $\text{pos}(n - 2) = (1, 3)$, $\text{pos}(n - 1) = (n - 3, 1)$: 2-block at axial distance $d_4 = c(n - 3, 1) - c(1, 3) = (4 - n) - 2 = 2 - n$. For $n \geq 8$, $|d_4| \geq 6$. $(T_{n-2}+1)v_{T_{(n-4,n-2)}^{B,B}} = (a_{d_4}+1)v_{T_{(n-4,n-2)}^{B,B}} + b_{d_4}v_{\rho_2}$, where $\rho_2 := s_{n-2}T_{(n-4,n-2)}^{B,B}$: $\text{pos}(n - 1) = (1, 3)$, $\text{pos}(n - 2) = (n - 3, 1)$. Apply T_{n-3} :

- On $v_{T_{(n-4,n-2)}^{B,B}}$: $\text{pos}(n - 3) = (n - 4, 1)$, $\text{pos}(n - 2) = (1, 3)$. 2-block at axial distance $d_5 = c(1, 3) - c(n - 4, 1) = 2 - (5 - n) = n - 3$, with $|d_5| \geq 5$ at $n \geq 8$. $(T_{n-3}+1)v_{T_{(n-4,n-2)}^{B,B}} = (a_{d_5}+1)v_{T_{(n-4,n-2)}^{B,B}} + b_{d_5}v_{\rho_3}$, $\rho_3 := s_{n-3}T_{(n-4,n-2)}^{B,B}$: $\text{pos}(n - 3) = (1, 3)$, $\text{pos}(n - 2) = (n - 4, 1)$.
- On v_{ρ_2} : $\text{pos}(n - 3) = (n - 4, 1)$, $\text{pos}(n - 2) = (n - 3, 1)$. Same column, adjacent. $(T_{n-3}+1)v_{\rho_2} = 0$.

Contribution from (B4): $(a_{d_4}+1)(a_{d_5}+1)v_{T_{(n-4,n-2)}^{B,B}} + (a_{d_4}+1)b_{d_5}v_{\rho_3}$.

(B5) $(a, b) = (n - 3, n - 2)$. $\text{pos}(n - 3) = (1, 2) = a$, $\text{pos}(n - 2) = (1, 3) = b$: same row, adjacent. Thus $T_{n-3}v_{T_{(n-3,n-2)}^{B,B}} = qv$ and $(T_{n-3}+1)v = (q + 1)v_{T_{(n-3,n-2)}^{B,B}}$. $\text{pos}(n - 2) = (1, 3)$, $\text{pos}(n - 1) = (n - 3, 1)$: 2-block at axial distance $d_4 = 2 - n$ (same as B4). $(T_{n-2}+1)v_{T_{(n-3,n-2)}^{B,B}} = (a_{d_4}+1)v_{T_{(n-3,n-2)}^{B,B}} + b_{d_4}v_{\rho'_2}$, where $\rho'_2 := s_{n-2}T_{(n-3,n-2)}^{B,B}$: $\text{pos}(n - 1) = (1, 3)$, $\text{pos}(n - 2) = (n - 3, 1)$.

Apply T_{n-3} :

- On $v_{T_{(n-3,n-2)}^{B,B}}$: same-row pair, so $(T_{n-3}+1)v = (q + 1)v$.
- On $v_{\rho'_2}$: $\text{pos}(n - 3) = (1, 2)$, $\text{pos}(n - 2) = (n - 3, 1)$. 2-block at axial distance $d_3 = 3 - n$ (same as the corresponding A -case). $(T_{n-3}+1)v_{\rho'_2} = (a_{d_3}+1)v_{\rho'_2} + b_{d_3}v_{\rho'_3}$, $\rho'_3 := s_{n-3}\rho'_2$: $\text{pos}(n - 3) = (n - 3, 1)$, $\text{pos}(n - 2) = (1, 2)$, $\text{pos}(n - 1) = (1, 3)$.

Contribution from (B5): $(q + 1)(a_{d_4}+1)v_{T_{(n-3,n-2)}^{B,B}} + b_{d_4}[(a_{d_3}+1)v_{\rho'_2} + b_{d_3}v_{\rho'_3}]$.

5.1 Disjointness of the five post-image supports

The five SYTs are tabulated by their values at $(1, 2)$ and $(1, 3)$:

SYT	(1, 2)	(1, 3)
$T_{(n-4, n-2)}^{B, B}$	$n - 4$	$n - 2$
$\rho_3 = s_{n-3} T_{(n-4, n-2)}^{B, B}$	$n - 4$	$n - 3$
$T_{(n-3, n-2)}^{B, B}$	$n - 3$	$n - 2$
$\rho'_2 = s_{n-2} T_{(n-3, n-2)}^{B, B}$	$n - 3$	$n - 1$
$\rho'_3 = s_{n-3} \rho'_2$	$n - 2$	$n - 1$

All five $((1, 2), (1, 3))$ values are distinct, so the five SYTs are distinct.

Proposition 10 (Lemma B). $\pi^{(c_2)} F_B \neq 0$. The seminormal expansion of $\pi^{(c_2)} F_B$ has five distinct supports $\{T_{(n-4, n-2)}^{B, B}, \rho_3, T_{(n-3, n-2)}^{B, B}, \rho'_2, \rho'_3\}$, each with coefficient in $\mathbb{Q}(q)^\times$.

Proof. Each contribution above involves a product of nonzero factors τ_B , the relevant $B_{(a, b)}$ (nonzero by Theorem 3), $(q + 1)$, and Hoefsmit coefficients $a_{d_i} + 1$ or b_{d_i} (nonzero by (1) since $|d_i| \geq 5$ at $n \geq 8$). The five SYTs are distinct (table above), so the five coordinates in $\pi^{(c_2)} F_B$ are independent and all nonzero. $\square$

6 Main theorem

Proof of Theorem 4. By May-20 Theorem 6.4 (the upper bound), $\dim B_{n-2}^{(\lambda)} V_\lambda \leq 2$, and the reduction formula (Theorem 1) shows $B_{n-2}^{(\lambda)} V_\lambda$ is spanned by $\{F_A, F_B\}$.

For the lower bound, suppose $c_A F_A + c_B F_B = 0$ for $c_A, c_B \in \mathbb{Q}(q)$. Apply $\pi^{(c_1)}$: $c_A \pi^{(c_1)} F_A + c_B \pi^{(c_1)} F_B = 0$. By Corollary 8, $\pi^{(c_1)} F_B = 0$, so $c_A \pi^{(c_1)} F_A = 0$. By Proposition 9, $\pi^{(c_1)} F_A \neq 0$, hence $c_A = 0$.

Symmetrically, applying $\pi^{(c_2)}$ and using Proposition 10 gives $c_B = 0$. Therefore $\{F_A, F_B\}$ is linearly independent and $\dim B_{n-2}^{(\lambda)} V_\lambda = 2$. $\square$

7 Computational verification

All checks at $q = 7$ in $\mathbb{Q}$. Scripts at `~/projects/scratch/2026-05-12-dim-2-321/`.

Direct dim and projection-witness check at $n \in \{8, 9, 10\}$. For each n :

- Compute $F_A, F_B \in V_\lambda$ via the three-branch reduction formula.
- Verify $\dim \text{span}(F_A, F_B) = 2 = \dim B_{n-2}^{(\lambda)} V_\lambda$ (both via direct $R'_{n-2} R'_{n-1} R'_n$ on V_λ).
- Verify $\text{supp}(F_A)$ contains only SYTs with $\text{pos}(n) \in \{(1, 3), (n - 3, 1)\}$ (0-indexed: $(0, 2)$ or $(n - 4, 0)$).
- Verify $\text{supp}(F_B)$ contains only SYTs with $\text{pos}(n) \in \{(2, 2), (n - 3, 1)\}$.
- Verify $\pi^{(c_1)} F_A$ has exactly 4 nonzero coordinates, on SYTs with $((1, 2), (2, 2)) \in \{(n - 4, n - 2), (n - 4, n - 3), (n - 3, n - 2)\}$ — matching the predictions in Proposition 9.
- Verify $\pi^{(c_2)} F_B$ has exactly 5 nonzero coordinates, on SYTs with $((1, 2), (1, 3)) \in \{(n - 4, n - 2), (n - 4, n - 3), (n - 3, n - 2)\}$ — matching Proposition 10.

All checks pass uniformly. Histograms: $|\text{supp}(F_A)|$ splits as 11 at $(n - 3, 1)$ / 4 at $(1, 3)$; $|\text{supp}(F_B)|$ splits as 13 at $(n - 3, 1)$ / 5 at $(2, 2)$.

8 Discussion

The structural picture. This is the first proof of a rank-corner formula at $r \geq 2$. The position-of- n invariant is the conceptual core: under the staircase $B_{\ell+1}^{(\lambda)} = R'_{\ell+1} \cdots R'_n$, the last operator R'_n uses T_{n-1} which moves n , but $R'_{\ell+1}, \dots, R'_{n-1}$ all use only T_j for $j \leq n-2$ and so preserve the position of n . Hence the position- n profile of the image after R'_n is preserved through the rest of the staircase. For λ with r length-preserving corners, $R'_n V_\lambda = E_{n-1}^+$ decomposes by $T_{n-1} + q$ -eigenvectors at the various corner pairs $\{c_i, c_k\}$ involving the column-1 bottom c_k , each carrying a different position of n at $c_i \neq c_j$ for different i . The position- n projections separate the contributions.

Conjecture: rank-corner formula in general. We extract from the proof the following:

Conjecture 11 (Rank-corner formula). *For any rank-zero $\lambda \vdash n$ that is not the sign representation, with r length-preserving corners and one column-1 bottom corner,*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = r.$$

The structural reason: the May-19 abstract Phase A theorem + multi-branch reduction yields Φ_X pieces for each length-preserving corner c_X paired with the column-1 bottom c_k . Each non-trivial pair contributes a 1-dimensional image in E_1^- ; pairs of length-preserving corners with each other contribute 0 via j -formula leakage (May-20 Lemma 4.1). The position-of- n invariant separates the r contributions *provided* the placement cells $\{c_1, \dots, c_{r-1}\}$ are pairwise distinct (which they always are by definition of corners). The non-vanishing of each piece is a finite Hoefsmit case analysis analogous to Sections 4, 5 of this paper.

Next steps.

- (1) Prove Conjecture 11 for $(4, 2, 1^{n-6})$ and $(5, 2, 1^{n-7})$ ($r = 2$) as further evidence; the technique in this paper should adapt verbatim (different axial distances but same case structure).
- (2) Handle $r = 3$: the smallest cases are $(4, 3, 2, 1^{n-9})$ and $(3, 3, 2, 1^{n-8})$ (corners at row 1, 2, 3, and column-1 bottom). The position-of- n argument generalizes; the case analysis grows but stays finite.
- (3) Connect to the underlying combinatorial object: the length-preserving corners of λ index “cell-pair states” in E_{n-1}^+ , and the rank-corner formula counts how many of these survive both the j -formula collapse and the $\text{im} \subseteq E_1^-$ constraint. A bijective description in terms of partition data would close the rank-zero programme.

Status. Theorem 4 closes the dimension gap flagged in May-20 §7 Remark 6.5(1). Combined with May-20 Theorem 6.4 and the May-19 abstract Phase A, $B_{\ell+1}^{(3,2,1^{n-5})} V_{(3,2,1^{n-5})}$ is now fully characterized: E_1^- containment + exact dimension 2.